

Exam Retake Scheduler

Capacity Aware Exam Retake
Scheduling System

USER MANUAL
& APPLICATION GUIDE

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OVERVIEW

Exam Retake Scheduler

The Exam Retake Scheduler is a web-based booking system designed to manage administrator-created exam retake sessions. The system supports capacity limits, scheduled time ranges, and multiple student bookings within each session.

Administrators can create retake sessions based on exam timing and academic requirements. The system facilitates communication between students, instructors, and administrators by providing a centralized and structured scheduling interface.

Students can book available sessions through public access without requiring an account, simplifying the booking process and improving accessibility.

System Requirements

Software Requirements

Web browser (Chrome, Microsoft Edge, Firefox, etc.)

Node.js (Frontend runtime environment)

Java (Backend server runtime)

Maven (for building and running the backend)

Hardware Requirements:

Standard personal computer or laptop

Internet browser capable of running modern web applications

Environment Requirements:

Local development environment

(application is not deployed publicly)

Ability to run both frontend and backend servers simultaneously

How to Run

These instructions describe how to run the Node.js (frontend) and Java/Maven (backend) environments used by the Exam Retake Scheduler.

Backend:

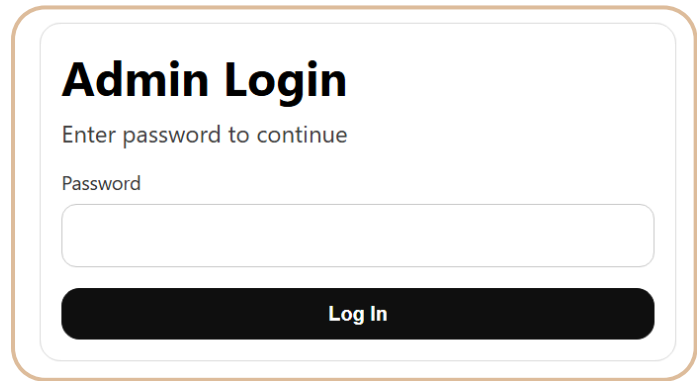
1. Navigate to the backend folder:
/code/backend
2. Run: `mvn compile exec:java`
3. The backend will run at:
`http://localhost:8081`

Frontend:

1. Navigate to the project folder:
/code
2. Run: `npm install`
3. Run: `npm run dev`
4. Open in browser:
`http://localhost:3000`

Both the backend and frontend must be running simultaneously for the application to function properly.

Admin Instructions

A screenshot of an 'Admin Login' form. The form has a title 'Admin Login' in bold. Below the title is a subtitle 'Enter password to continue'. There is a label 'Password' above a text input field. At the bottom of the form is a black button with the text 'Log In' in white.

How to Log In

Follow these steps to access the admin dashboard.

1. Go to /admin/login
2. Enter the admin password
3. Click “Log In”

How to Create a Session Slot

1. Navigate to the Create Slots tab
2. Select a department from the dropdown
3. Enter:
 - Session name
 - Date
 - Capacity
 - Start time
 - End time
4. Click “Create Session”

How to View Created Slots

1. Go to the Created Slots tab
2. Click “Refresh Slots”
3. View all sessions for the currently selected department

How to Modify a Slot

1. In Created Slots, click “Edit” on a session
2. Modify fields as needed
3. Click “Save”

How to Delete a Slot

1. Click “Delete” on a session
2. Confirm the deletion

How to View Student Bookings

1. Go to the Booked Appointments tab
2. Click “Refresh Bookings”
3. View student information for each booking

Student Instructions

Student Dashboard

Department: Homeland Security

Available Sessions

No sessions available right now.

Refresh

Booking

Choose a session to open the booking form.

Student Dashboard

Department: Homeland Security

Available Sessions

Test 4
2026-04-29 • 09:00 - 11:00
5 of 5 seats left

Book this session

Booking

Selected: Homeland Security • 2026-04-29 • 09:00 - 11:00 Close

Book Appointment

Confirm your details below.

Department: **Homeland Security**

When: **2026-04-29 • 09:00 - 11:00**

Where: **TBD**

Full name *

Email *

Student ID (optional)

Exam Type *

☒ MWF (50 min) ☐ TTH (75 min)

Course code *

Instructor Email *

Notes (optional)

Confirm booking

Booking a Session

Follow these steps to book a retake session:

1. Navigate to the student page
2. Select a department
3. View available sessions
4. Choose a session
5. Enter:
 - Name
 - Email
 - Course code
 - Instructor
 - Exam type
6. Submit the form

Notes & Limitations

The following reflect the current scope of the system:

- Students book a retake session as a whole rather than selecting a specific time within that session.
- Retake sessions must be created individually; recurring sessions are not currently supported.
- Administrator access is protected by a simple password system intended for demonstration purposes.
- Data is stored locally and is not connected to a persistent database.
- Bookings are automatically confirmed upon submission.
- Administrator approval (confirm/reject workflow) was considered but is not implemented in the current version of the system.

Reach out!



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